

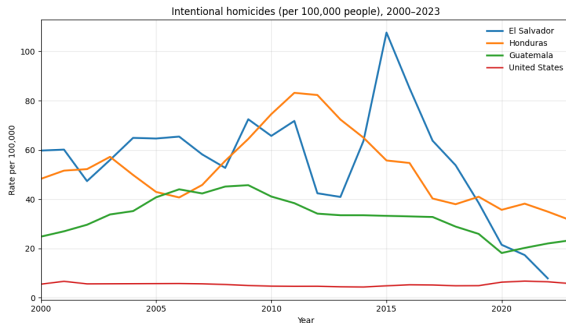
Being Tough on Crime: Effects on School Dropout

Francesco Sbrana

March 24, 2026

Motivation

- **Crime** is a widespread problem in Latin American countries
- Mostly perpetrated by criminal groups having the form of **Street Gangs**



Source: World Bank (UNODC)

"Mano Dura" policies

Since 2000s, governments routinely resort to "**Mano Dura**" policies to **crack-down** extortion and violence. (Cutrona et al., 2025)

General characteristics:

- Suspension of constitutional rights
- Use of military
- Arbitrary detentions
- Supported by the population (Rosen, 2021) (Rodgers et al., 2009)

Honduras' State of Exception

Honduras introduced a new "Mano Dura" policy in **December 2022**, with the main objective of reducing extortion and dismantle gangs

Honduras' State of Exception

		
DIARIO OFICIAL DE LA REPÚBLICA DE HONDURAS		
La primera imprenta llegó a Honduras en 1829, siendo instalada en Tegucigalpa, en el cuartel San Francisco, lo primero que se imprimió fue una proclama del General Morazán, con fecha 4 de diciembre de 1829.	 EMPRESA NACIONAL DE ARTES GRÁFICAS E.N.A.G.	Después se imprimió el primer periódico oficial del Gobierno con fecha 25 de mayo de 1830, conocido hoy, como Diario Oficial "La Gaceta".
AÑO CXLIV TEGUCIGALPA, M. D. C., HONDURAS, C. A.		SÁBADO 3 DE DICIEMBRE DEL 2022. NUM. 36,093
Sección A		
<u>Poder Ejecutivo</u>	<u>SUMARIO</u>	
DECRETO EJECUTIVO NÚMERO PCM 29-2022	Sección A	
	Decretos y Acuerdos	
	PODER EJECUTIVO	
	Decreto Ejecutivo Número PCM 29-2022	A. 1 - 10
	OTROS	A. 11
	AVANCE	A. 12
LA PRESIDENTA CONSTITUCIONAL DE LA REPÚBLICA EN CONSEJO DE SECRETARIOS DE ESTADO,	Sección B	
	Avisos Legales	B. 1 - 8
	Desprendible para su comodidad	

Honduras' State of Exception

DECRETA:

ARTÍCULO 1. En virtud de la grave perturbación de la paz y la seguridad que prevalece en las principales ciudades del país ocasionada esencialmente por grupos criminales organizados, que operan como mafias poniendo en riesgo la vida y los bienes de las personas, incurriendo y consumando delitos de extorsión, asesinatos, robos, tráfico de drogas y secuestros, que reclama el pueblo como alarmante calamidad pública, se resuelve: suspender las garantías establecidas en la Constitución de la República en los artículos 69, 78, 81, 84, 93 y 99 a partir de las 6:00 p. m. del día martes 6 de diciembre del 2022 y hasta las 6:00 p. m. del día jueves 6 de enero del año 2023. En consecuencia, a la Secretaría de Estado en el Despacho de Seguridad Pública, a través de la Policía Nacional Preventiva respetando el principio de necesidad y proporcionalidad necesarios para mantener la seguridad el orden y la paz en la comunidad, se le faculta para detener a las personas que determine y considere responsables de asociarse, ejecutar, o tener vinculaciones, en la comisión de delitos y crímenes contemplados en este Decreto, en los sectores de los municipios del Distrito Central y de San Pedro

de Estado en el Despacho de Defensa Nacional, las Fuerzas Armadas y la Policía Militar del Orden Público (PMOP), coordinarán acciones para dar cumplimiento al presente Decreto y mantener el orden, la paz y la seguridad nacional, así como el control de las fronteras terrestres, aéreas y marítimas.

ARTÍCULO 4. Las autoridades policiales competentes al momento de la detención deberán identificarse, informar los motivos de la detención y respetar los derechos de los detenidos. En los centros de detención deberá llevarse un registro oficial de los detenidos conforme los estándares internacionales.

ARTÍCULO 5. Se convoca al Congreso Nacional para que dentro del plazo de treinta (30) días, conozca del presente Decreto y lo ratifique, modifique o impruebe.

ARTÍCULO 6. Se instruye a la Secretaría de Estado en el Despacho de Relaciones Exteriores y Cooperación Internacional informar inmediatamente el presente Decreto una vez aprobado, a los demás Estados Parte de la Convención

Honduras' State of Exception

Honduras introduced a new "Mano Dura" policy in **December 2022**, with the main objective of reducing extortion and dismantle gangs

Suspended rights include:

- Protection against arbitrary detention
- Due process safeguards
- Freedom of movement
- Freedom of assembly
- Protection against warrantless home searches

Honduras' State of Exception



Criticisms

- No evidence of crime reduction (Escaño et al., 2025) (Gutiérrez Rivera, 2011)
- Dramatic increase in incarceration rates (Santos et al., 2026)
- Undermine democratic values (Blair et al., 2025) (Holland, 2013) (Hume, 2007) (Breda, 2023)
- Evidence of human right abuses attributed to security forces
- Gangs adaptation (Rodgers et al., 2009)

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No research on socio-economic consequences of Mano Dura policies implementations, yet.

Research Question

"What is the effect of "Mano Dura" on school dropout?"

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Methodology: Difference-in-Differences

Exploiting the spatial (and staggered) rollout of the policy

Honduras' Context

Education: (World Bank, 2020) (Adelman et al., 2017)

- 9.6 average years of schooling
- $\approx 65\%$ drop school between sixth and tenth grade [Show](#)

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Education: (World Bank, 2020) (Adelman et al., 2017)

- 9.6 average years of schooling
- $\approx$ 65% drop school between sixth and tenth grade [Show](#)

Gangs Recruitment: (Cruz et al., 2020)

- Most "Maras" are youth [Show](#)
- $\approx$ 50% join gangs between 12 and 16 yo [Show](#)
- $\approx$ 50% don't arrive to middle school [Show](#)
- $\approx$ 80% commit first offense before turning 18 ($\approx$ 32% before 14) [Show](#)
- $\approx$ 60% have the first child before 20yo [Show](#)
- $\approx$ 65% join gang because peer pressure or to hang out with peers [Show](#)

Possible Effects on School Dropout

Income Effect ↑

Security Effect ↓

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- Arrests increase
- Lack of breadwinners
- Child might be forced to leave school
- Possibly go working, either in the formal or informal (crime) sector

Security Effect ↓

- Security increases inside treated municipalities and near schools
- Less “barriers” in accessing education
(Melnikov et al., 2020)
- Less gang recruitment nearby schools

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Security Effect ↓

- Security increases inside treated municipalities and near schools
- Less “barriers” in accessing education (Melnikov et al., 2020)
- Less gang recruitment nearby schools

- Expect heterogenous effects across **Gender** and **Grades** (Castro et al., 2025)

Literature Review

- **Mano Dura, anti-gang policy, and criminal governance**

Why governments adopt hardline anti-crime policies, how gangs adapt, and the political consequences of repression.

(Holland, 2013) (Cutrona et al., 2025) (Gutiérrez Rivera, 2011) (Muggah et al., 2018) (Santos et al., 2026) (Hume, 2007)

- **Education, crime, and criminal organizations**

Violence, gangs, and organized crime affect schooling through fear, stress, extortion, recruitment, and reduced mobility.

(Lochner, 2010) (Aizer, 2008) (Michaelsen et al., 2020) (Melnikov et al., 2020) (Sviatschi, 2022) (Adelman et al., 2017)

- **Education and law enforcement**

How policing and punitive exposure affect absenteeism, discipline, attainment, and dropout.

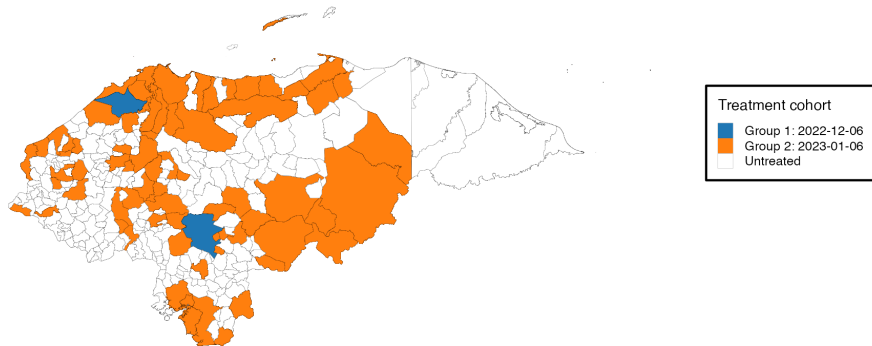
(Ang, 2021) (Weisburst, 2019) (Fisher et al., 2023)

Setting

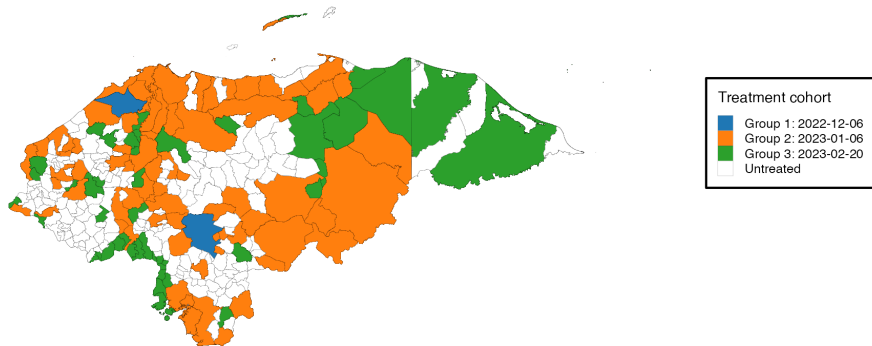
Treatment Rollout



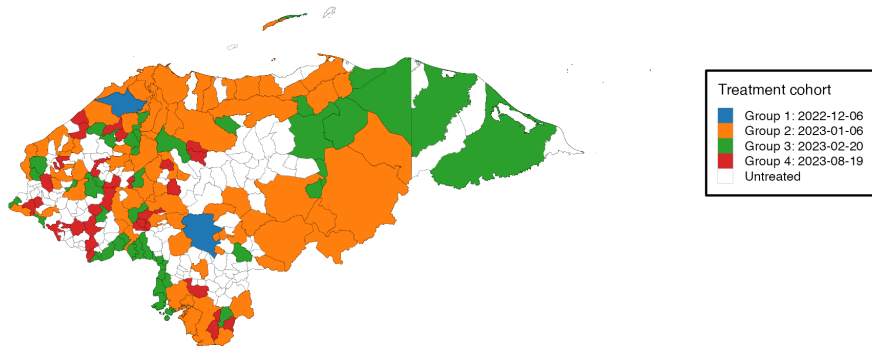
Treatment Rollout



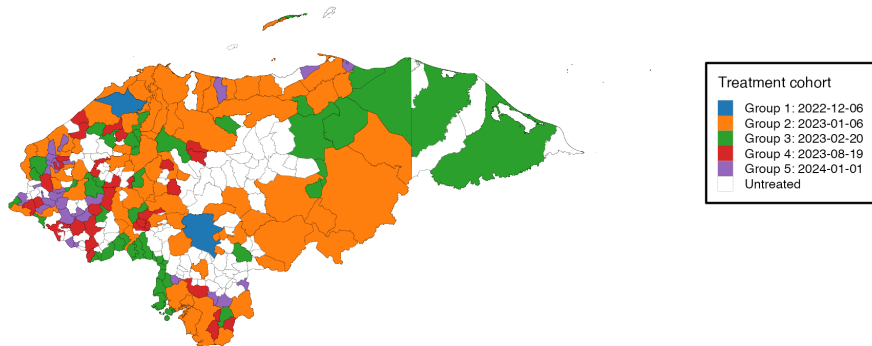
Treatment Rollout



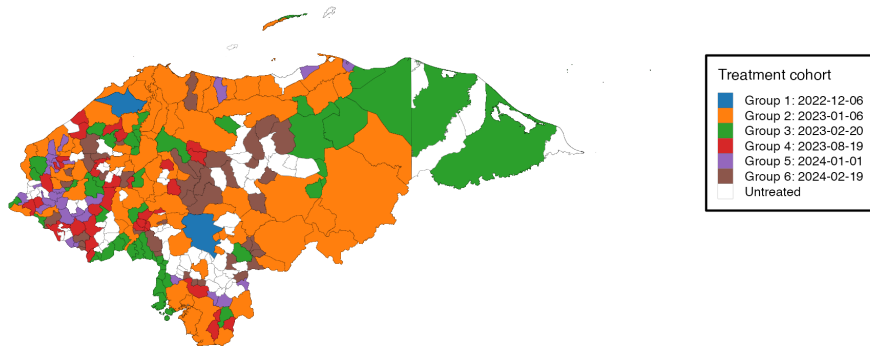
Treatment Rollout



Treatment Rollout



Treatment Rollout



225 Treated and **73** Control Municipalities

Data

Obtained

- **Treatment** data: date, municipalities affected
- **School** data (Unbalanced Panel)
 - Initial and Final enrollement per year (2019 - 2025)
 - Schools IDs, Locations, Levels (primary, secondary, etc.), Age of students
 - School characteristics (Public/Private, Rural/Urban)

To be requested

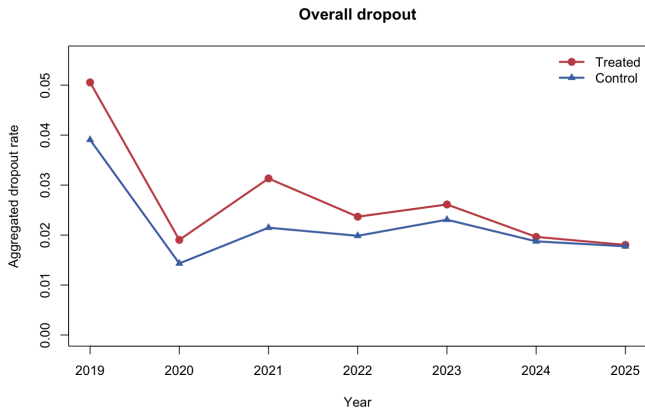
- **School** Characteristics
- **Municipality** Characteristics
- **Student** Data (Attendance, Marks)
- **Displacement** Data
- **Administrative Crime** Data

Outcome Definition

$$\text{Dropout rate}_{it} = \frac{\sum_i \# \text{ Dropout}_{it}}{\sum_i \# \text{ Initial Registration}_{it}}$$

where units i are *Schools* or *Municipalities* depending on the level of analysis.

Outcome Evolution - Overall Dropout



Male Dropout

Female Dropout

Summary Statistics

Table: Summary Statistics, Balanced School-Level Data, Group 3 and Never Treated

Variable	Treated				Control			
	Mean	SD	Min	Max	Mean	SD	Min	Max
Dropout rate	0.018	0.047	0.000	0.632	0.012	0.036	0.000	0.444
Dropout rate (female)	0.017	0.055	0.000	0.750	0.011	0.039	0.000	0.500
Dropout rate (male)	0.019	0.060	0.000	1.000	0.014	0.050	0.000	1.000
Government share	0.977	0.150	0.000	1.000	0.991	0.096	0.000	1.000
Rural share	0.903	0.295	0.000	1.000	0.898	0.303	0.000	1.000
Female student share	0.498	0.124	0.000	1.000	0.495	0.124	0.000	1.000
Initial enrollment	58.909	94.713	1.000	1095.000	48.319	73.690	1.000	1077.000
Observations	2,929				2,599			

Notes: This table reports school-level summary statistics for treated and control schools in 2022, for Group 3 and Never Treated.

Raw School

Raw Municipality

Balanced School

Balanced Municipality

Group 3 Municipality

Balance Table

Table: Balance Table, School-Level Data, Group 3 and Never Treated

Variable	Treated Mean	Control Mean	Difference	p-value	Std. Diff.
Dropout rate	0.018	0.012	0.005	0.001	0.129
Dropout rate (female)	0.017	0.011	0.006	0.001	0.134
Dropout rate (male)	0.019	0.014	0.005	0.001	0.092
Government share	0.977	0.991	-0.014	0.001	-0.109
Rural share	0.903	0.898	0.005	0.508	0.018
Female student share	0.498	0.495	0.003	0.407	0.022
Initial enrollment	58.909	48.319	10.590	0.001	0.125
N	2,929	2,599			

Notes: This table reports covariate balance for the school-level subsample in 2022. Difference is defined as treated mean minus control mean. Std. Diff. denotes the standardized difference in means.

Balance Municipality

Preliminary Results

Econometric Specifications

- **Baseline TWFE**

$$Y_{it} = \alpha_i + \delta_t + \theta (T_i \times \text{Post}_t) + \varepsilon_{it}$$

- **Dynamic Event-Study**

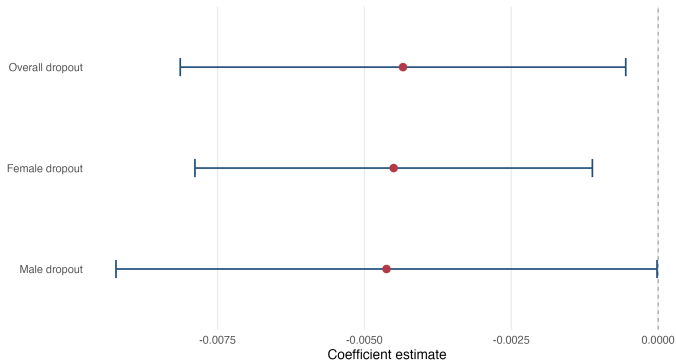
$$Y_{it} = \alpha_i + \delta_t + \sum_{\substack{j=-J \\ j \neq -1}}^K \theta_j (T_i \times \mathbf{1}\{t = j\}) + \varepsilon_{it}$$

Where:

- Y_{it} : dropout outcome
- α_i : unit fixed effects
- δ_t : year fixed effects
- Post_t : post-2023 period
- SE clustered at municipality level

TWFE Results - School Panel

TWFE coefficients: school-year panel



TWFE with unit FE, year FE, and municipality-clustered standard errors.

Municipality

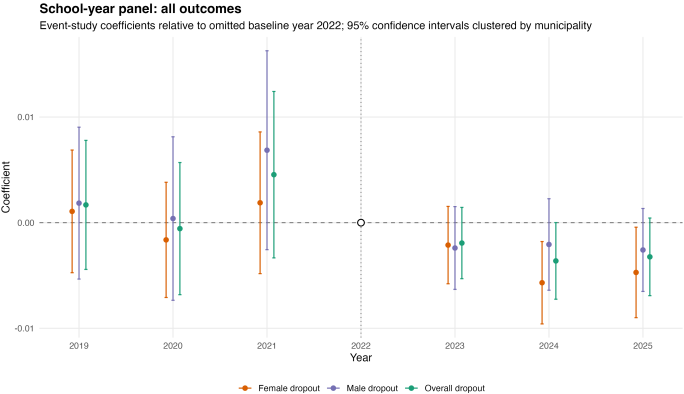
TWFE estimates, school-year panel

	Overall dropout	Female dropout	Male dropout
Treatment ($group_3 \times post_{2023}$)	-0.0043** (0.0019)	-0.0045** (0.0017)	-0.0046* (0.0023)
Observations	38,693	38,532	38,536
Unit FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Clustered SE	Municipality	Municipality	Municipality

Notes: SE clustered at the Municipality Level. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Municipality

Event Study Results - School Panel



Overall School Male School Female School

Municipality Coefficients

Conclusions and Next Steps

Results suggest overall negative effects of Mano Dura policies on school dropout (-0.45 pp), with more credible effects for female students. Identification needs to be strengthened.

Next steps:

- Expand pre-periods
- Add Covariates
- Heterogeneity Across School Grades
- Explore Violence Changes
- Explore Displacement

Thank you!

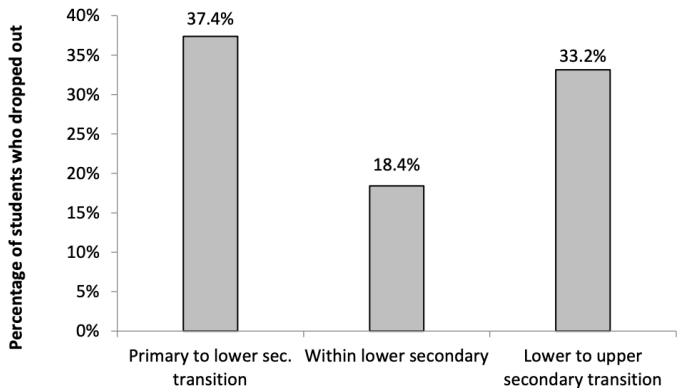
Feedbacks are welcome :)

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Appendix

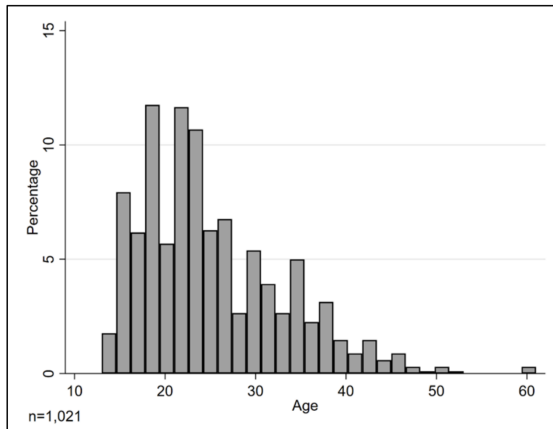
Dropout levels Honduras

Figure 3. Dropout rates in Honduras



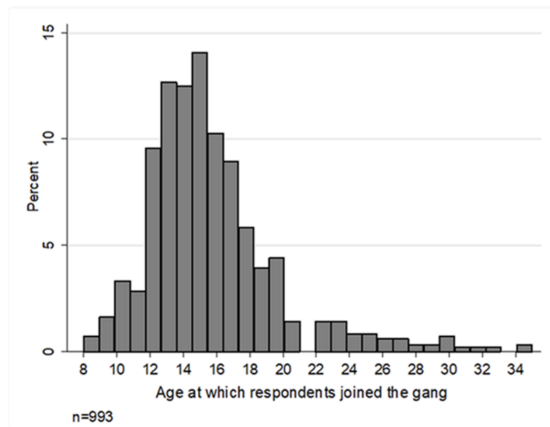
Age Distribution of Gang Members

Exhibit 4. Age Distribution Among Respondents



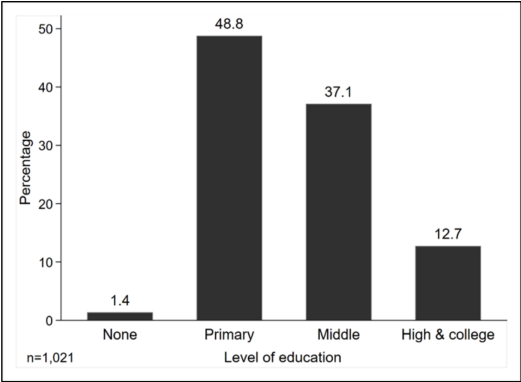
Age when joining gang

Exhibit 19. Age When Joining a Gang in Honduras



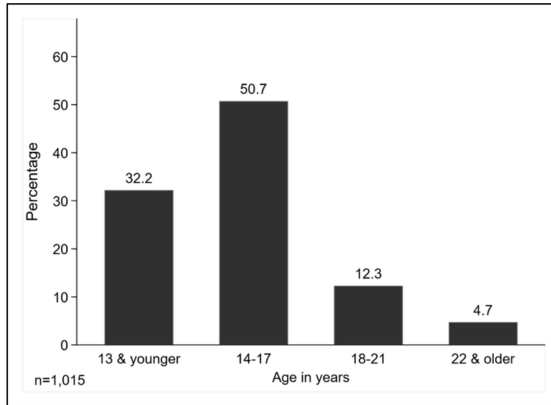
Educational Level of Gang Members

Exhibit 5. Level of Education Among Gang Members



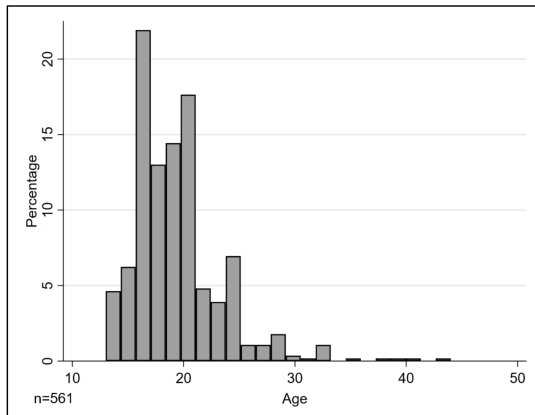
Age of First Offense

Exhibit 12. Age of First Offense



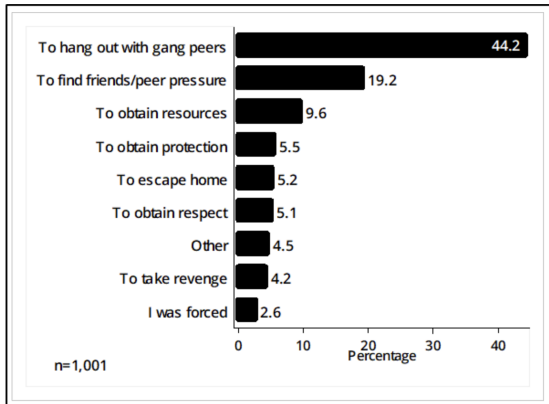
Age of Having First Child

Exhibit II. Age of Having First Child

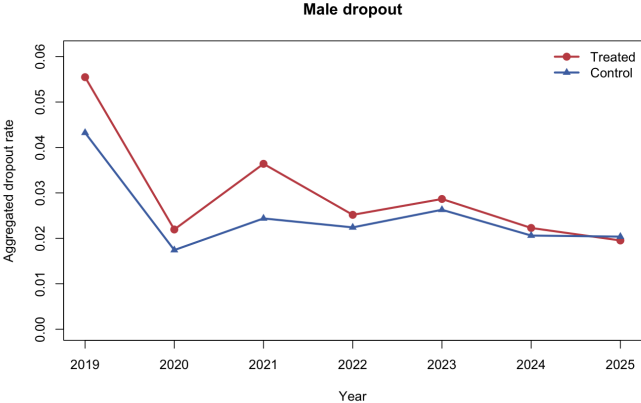


Reasons for Joining a Gang

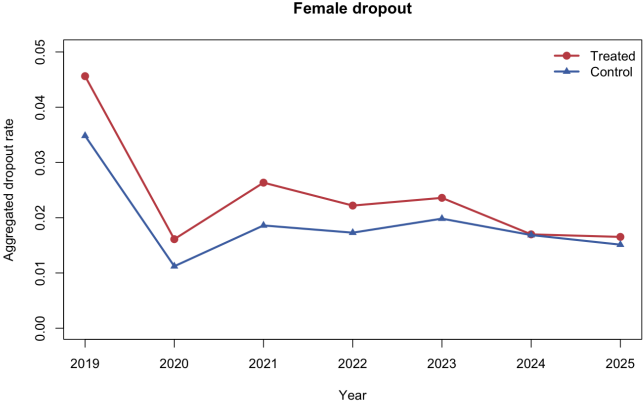
Exhibit 21. Reasons for Joining the Gang



Outcome Evolution - Male



Outcome Evolution - Female



Summary Statistics

Table: Summary Statistics, Raw School-Level Data (Unbalanced)

Variable	Treated				Control			
	Mean	SD	Min	Max	Mean	SD	Min	Max
Dropout rate	0.020	0.050	0.000	1.000	0.013	0.038	0.000	0.444
Dropout rate (female)	0.018	0.050	0.000	1.000	0.011	0.043	0.000	0.667
Dropout rate (male)	0.023	0.062	0.000	1.000	0.014	0.052	0.000	1.000
Government share	0.915	0.280	0.000	1.000	0.989	0.105	0.000	1.000
Rural share	0.792	0.406	0.000	1.000	0.901	0.298	0.000	1.000
Female student share	0.498	0.125	0.000	1.000	0.497	0.130	0.000	1.000
Initial enrollment	79.544	167.712	0.000	6051.000	45.546	71.464	0.000	1077.000
Observations	21,582				2,868			

Notes: This table reports raw school-level summary statistics for treated and control schools in 2022.

Summary Statistics

Table: Summary Statistics, Raw Municipality-Level Data (Unbalanced)

Variable	Treated				Control			
	Mean	SD	Min	Max	Mean	SD	Min	Max
Dropout rate	0.026	0.015	0.000	0.089	0.021	0.016	0.000	0.076
Dropout rate (female)	0.023	0.014	0.000	0.085	0.018	0.014	0.000	0.070
Dropout rate (male)	0.029	0.017	0.000	0.093	0.024	0.019	0.000	0.082
Government share	0.962	0.071	0.561	1.000	0.982	0.064	0.556	1.000
Rural share	0.857	0.120	0.264	0.972	0.857	0.149	0.000	0.976
Female student share	0.505	0.016	0.436	0.543	0.499	0.017	0.453	0.537
Initial enrollment	7629.898	20378.746	266.000	251034.000	1789.397	1108.093	316.000	6607.000
Observations	225				73			

Notes: This table reports raw municipality-level summary statistics for treated and control municipalities in 2022.

Summary Statistics

Table: Summary Statistics, Balanced School-Level Data

Variable	Treated				Control			
	Mean	SD	Min	Max	Mean	SD	Min	Max
Dropout rate	0.021	0.048	0.000	1.000	0.012	0.036	0.000	0.444
Dropout rate (female)	0.019	0.050	0.000	1.000	0.011	0.039	0.000	0.500
Dropout rate (male)	0.023	0.060	0.000	1.000	0.014	0.050	0.000	1.000
Government share	0.926	0.263	0.000	1.000	0.991	0.096	0.000	1.000
Rural share	0.794	0.405	0.000	1.000	0.898	0.303	0.000	1.000
Female student share	0.498	0.116	0.000	1.000	0.495	0.124	0.000	1.000
Initial enrollment	85.570	175.091	0.000	6051.000	48.319	73.690	1.000	1077.000
Observations	19,008				2,599			

Notes: This table reports school-level summary statistics for treated and control schools in 2022.

Summary Statistics

Table: Summary Statistics, Balanced Municipality-Level Data

Variable	Treated				Control			
	Mean	SD	Min	Max	Mean	SD	Min	Max
Dropout rate	0.026	0.015	0.000	0.091	0.021	0.016	0.000	0.076
Dropout rate (female)	0.024	0.014	0.000	0.087	0.018	0.014	0.000	0.070
Dropout rate (male)	0.029	0.017	0.000	0.094	0.024	0.019	0.000	0.082
Government share	0.967	0.065	0.597	1.000	0.985	0.062	0.556	1.000
Rural share	0.852	0.121	0.270	0.970	0.853	0.149	0.000	0.980
Female student share	0.505	0.017	0.436	0.543	0.498	0.018	0.449	0.536
Initial enrollment	7228.978	19207.748	254.000	237807.000	1720.274	1070.096	316.000	6479.000
Observations	225				73			

Notes: This table reports municipality-level summary statistics for treated and control municipalities in 2022.

Summary Statistics

Table: Summary Statistics, Balanced Municipality-Level Data, Group 3 and Never Treated

Variable	Treated				Control			
	Mean	SD	Min	Max	Mean	SD	Min	Max
Dropout rate	0.023	0.014	0.006	0.071	0.021	0.016	0.000	0.076
Dropout rate (female)	0.021	0.014	0.005	0.070	0.018	0.014	0.000	0.070
Dropout rate (male)	0.025	0.015	0.004	0.071	0.024	0.019	0.000	0.082
Government share	0.980	0.045	0.768	1.000	0.985	0.062	0.556	1.000
Rural share	0.879	0.091	0.475	0.967	0.853	0.149	0.000	0.980
Female student share	0.503	0.020	0.442	0.538	0.498	0.018	0.449	0.536
Initial enrollment	3594.667	2777.975	254.000	16732.000	1720.274	1070.096	316.000	6479.000
Observations	48				73			

Notes: This table reports municipality-level summary statistics for treated and control municipalities in 2022, for Group 3 and Never Treated.

Balance Table

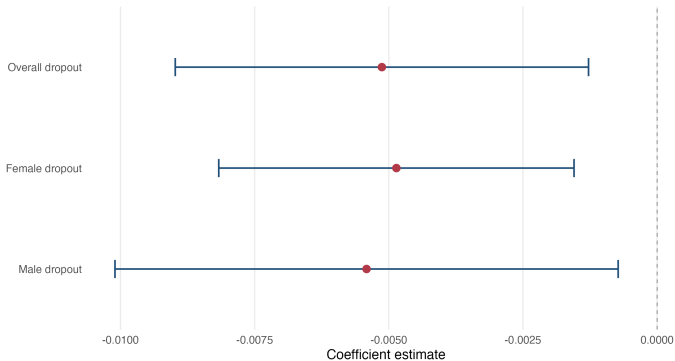
Table: Balance Table, Municipality-Level Data, Group 3 and Never Treated

Variable	Treated Mean	Control Mean	Difference	p-value	Std. Diff.
Dropout rate	0.023	0.021	0.002	0.442	0.141
Dropout rate (female)	0.021	0.018	0.003	0.201	0.239
Dropout rate (male)	0.025	0.024	0.001	0.749	0.058
Government share	0.980	0.985	-0.006	0.564	-0.104
Rural share	0.879	0.853	0.026	0.243	0.207
Female student share	0.503	0.498	0.005	0.194	0.245
Initial enrollment	3594.667	1720.274	1874.393	0.001	0.890
N	48	73			

Notes: This table reports covariate balance for the municipality-level subsample in 2022. Difference is defined as treated mean minus control mean. Std. Diff. denotes the standardized difference in means.

TWFE Results - Municipality Panel

TWFE coefficients: municipality-year panel



TWFE with unit FE, year FE, and municipality-clustered standard errors.

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TWFE Results

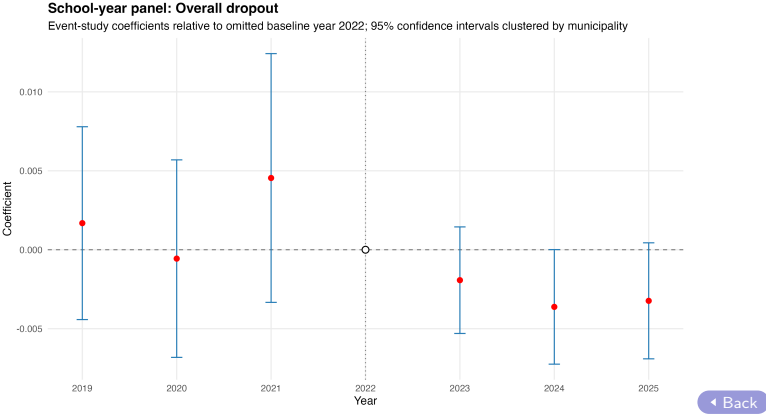
TWFE estimates, municipality-year panel

	Overall dropout	Female dropout	Male dropout
Treatment ($group_3 \times post_{2023}$)	-0.0051** (0.0020)	-0.0049*** (0.0017)	-0.0054** (0.0024)
Observations	847	847	847
Unit FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Clustered SE	Municipality	Municipality	Municipality

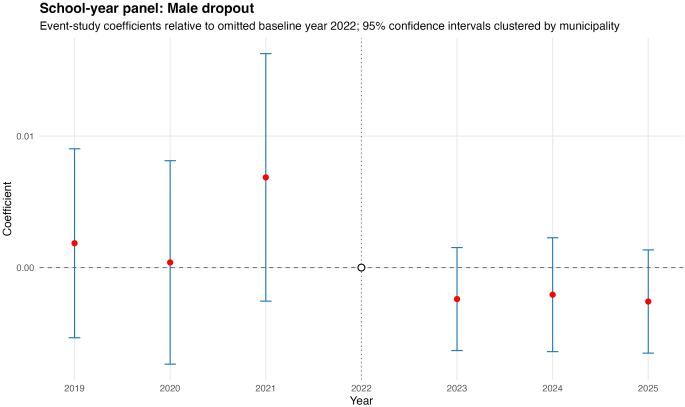
Notes: SE clustered at the Municipality Level. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

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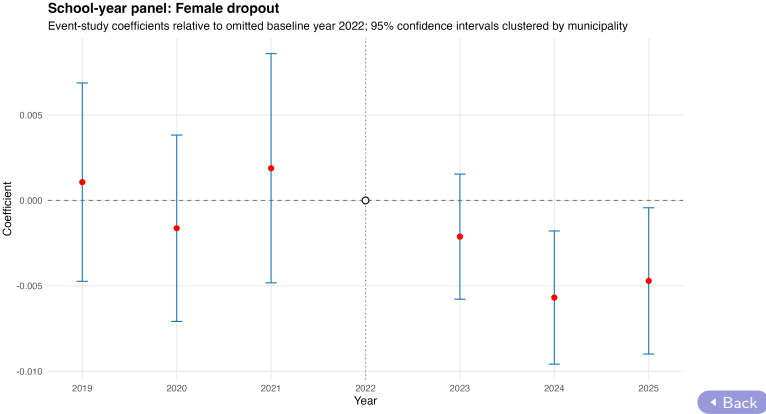
Event Study Results - School Panel - Overall Dropout



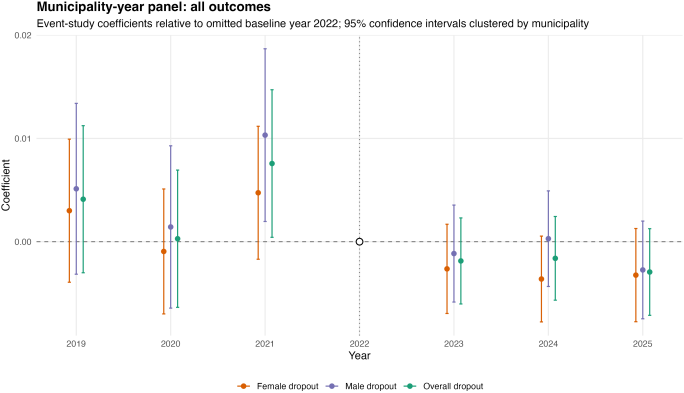
Event Study Results - School Panel - Male Dropout



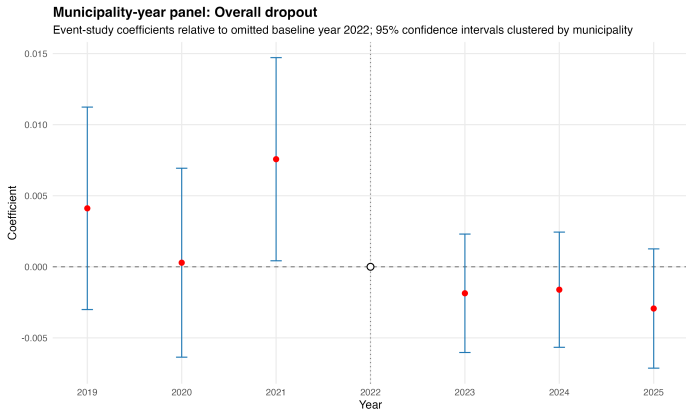
Event Study Results - School Panel - Female Dropout



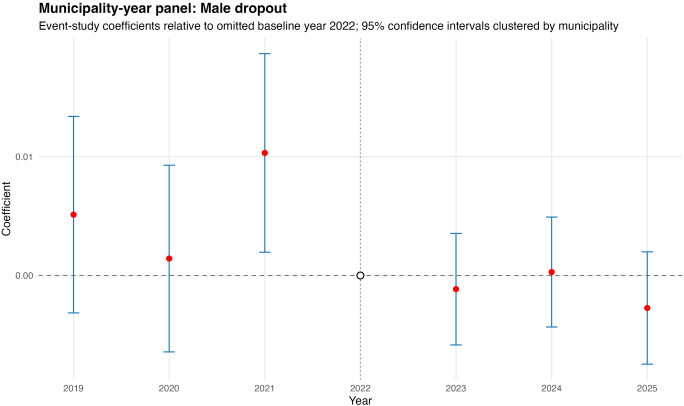
Event Study Results - Municipality Panel



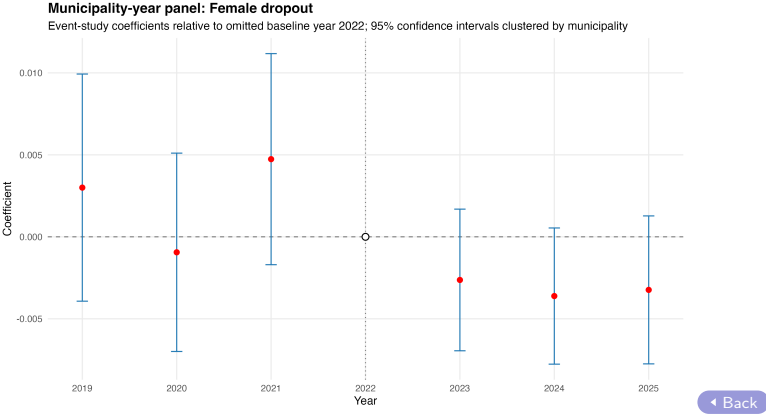
Event Study - Municipality Panel - Overall Dropout



Event Study - Municipality Panel - Male Dropout



Event Study - Municipality Panel - Female Dropout



Event Study Results

Event-study estimates for school-year panel

	Overall dropout	Female dropout	Male dropout
2019 ($e = -4$)	0.0017 (0.0031)	0.0011 (0.0030)	0.0019 (0.0037)
2020 ($e = -3$)	-0.0006 (0.0032)	-0.0016 (0.0028)	0.0004 (0.0039)
2021 ($e = -2$)	0.0045 (0.0040)	0.0019 (0.0034)	0.0069 (0.0048)
2023 ($e = 0$)	-0.0019 (0.0017)	-0.0021 (0.0019)	-0.0024 (0.0020)
2024 ($e = +1$)	-0.0036* (0.0019)	-0.0057*** (0.0020)	-0.0021 (0.0022)
2025 ($e = +2$)	-0.0032* (0.0019)	-0.0047** (0.0022)	-0.0026 (0.0020)
Observations	38,693	38,532	38,536
Unit FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Clustered SE	Municipality	Municipality	Municipality

Notes: Standard errors are clustered at ID.Municipio. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Event Study Results

Event-study estimates for municipality-year panel

	Overall dropout	Female dropout	Male dropout
2019 ($e = -4$)	0.0041 (0.0036)	0.0030 (0.0035)	0.0051 (0.0042)
2020 ($e = -3$)	0.0003 (0.0034)	-0.0009 (0.0031)	0.0014 (0.0040)
2021 ($e = -2$)	0.0076** (0.0036)	0.0047 (0.0033)	0.0103** (0.0043)
2023 ($e = 0$)	-0.0019 (0.0021)	-0.0026 (0.0022)	-0.0011 (0.0024)
2024 ($e = +1$)	-0.0016 (0.0021)	-0.0036* (0.0021)	0.0003 (0.0024)
2025 ($e = +2$)	-0.0029 (0.0021)	-0.0032 (0.0023)	-0.0027 (0.0024)
Observations	847	847	847
Unit FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Clustered SE	Municipality	Municipality	Municipality

Notes: Standard errors are clustered at ID.Municipio. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.